



Improving Histopathology Classification

How data deprivations affect discrimination and calibration

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Motivation

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Common histopathology problems



Class imbalance

Some classes occur far more often than others; real-world label frequencies are typically long-tailed rather than balanced.



Data shortage

A class can be short on patches, short on patients, or both — and these two kinds of shortage are not the same problem.



Difficulty variation

Some classes are intrinsically harder to distinguish than others, independent of how much data supports them.

Motivation

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Identify weaknesses in learning methods and try to improve them



GOAL

**Improve
discrimination and/
or calibration**



FOCUS

**Long-tailed
histopathology
patch classification**

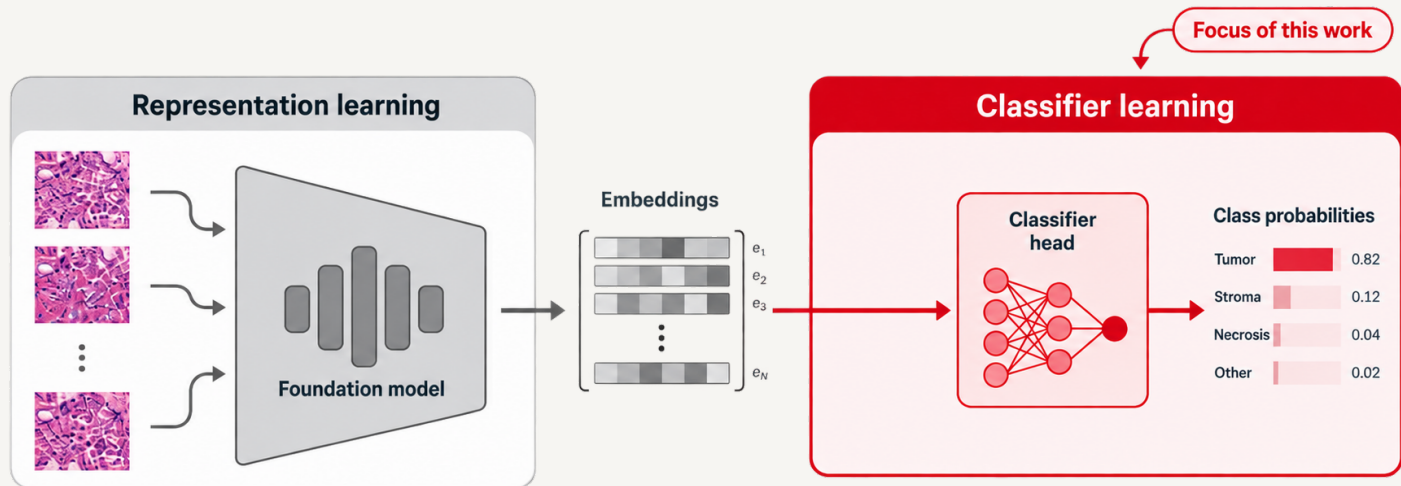
CORE QUESTION

Under what conditions do current learning methods fail?

Setting

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Classifier learning on fixed histopathology representations



WSI patches encoded by a frozen **Virchow2** foundation model

Small MLP classifier maps embeddings to class probabilities — where mitigation happens

✓ Patient-/case-disjoint evaluation

↗ Cleanly separates representation from classifier learning

⚠ Related benchmarks often mix interventions across both stages

Idea

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Induce different data deprivations and evaluate mitigation success

Problem	Question
Prevalence	Are classes unequally represented?
Support	Nominal — How many patches support each class? Distinct-unit — How many patients / slides provide that evidence?
Difficulty	Are scarce classes also intrinsically harder?
Diversity	How much variation is represented within a class?

Controlled Training Conditions

Constructing different types of data deprivation



Prevalence

Class-frequency ratio manipulated directly — balanced, ~10× moderate, and ~100× severe head-tail imbalance.



Nominal support

Total patch count per class varied at fixed prevalence, isolating raw patch volume from the imbalance ratio.



Distinct-unit support

Patch counts held fixed while patient/slide count drops — the patient-shortage condition decouples patches from patients.



Difficulty

Scarce classes assigned as difficulty-aligned, reversed, or native, separating intrinsic difficulty from prevalence.

Why no diversity condition?

- Diversity was measured in frozen Virchow2 space
- At these patch counts, changing which patches are selected barely changes semantic volume
- Diversity therefore remains **exploratory**, not independently manipulated

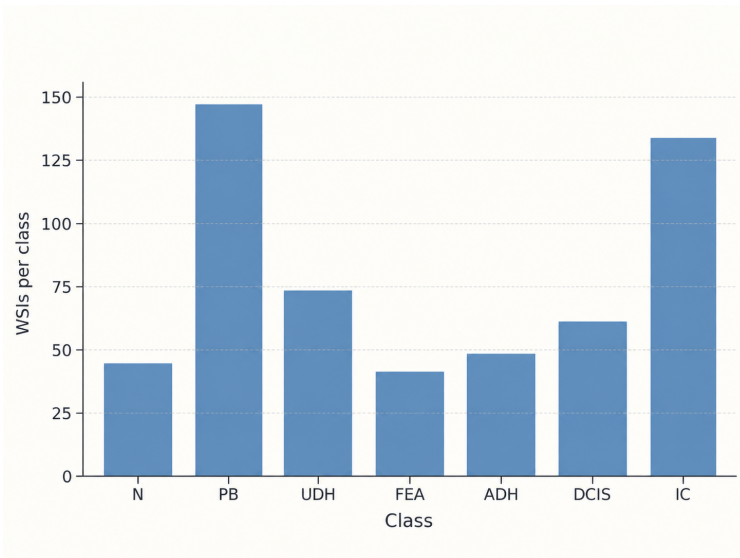
Datasets

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Multiclass histopathology classification

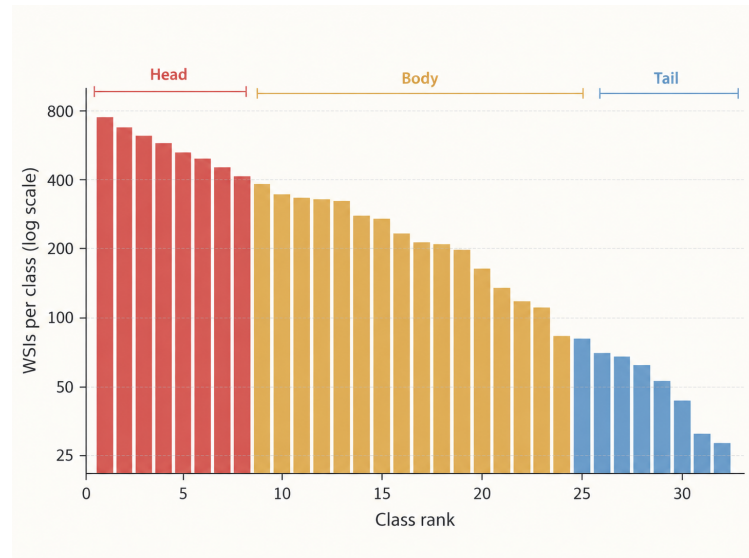
BRACS

Breast lesions · 7 classes



TCGA-UT

Unified tumour-type task · 30 classes



Methods

Does addressing the problem directly mitigate it best?

Class-weighted cross-entropy

Hold the mechanism fixed and vary only the signal

Signal	Weighted-CE variant
Prevalence	Class-frequency weighted CE
Nominal support	Class-balanced CE
Distinct-unit support	Distinct-unit-support weighted CE
Difficulty	Pilot-difficulty weighted CE
Diversity	Semantic-scale weighted CE

Wider roster

Popular long-tailed learning methods — tests whether weighting itself is too restrictive

Balanced sampling
Logit adjustment
cRT
Focal Loss
CFAL
Soft-F1
Soft-MCC
OKO

Results

Prevalence, nominal support and difficulty interventions have effects but are repairable

Signal-based Reweighting

Recovery (%)	BRACS	TCGA-UT
Prevalence	87.1	85.3
Nominal support	89.2	81.2
Independent support	19.7	47.7
Difficulty	16.4	2.8
Diversity	14.2	12.6

Wider Roster

Recovery (%)	BRACS	TCGA-UT
Balanced sampling	90.8	99.1
CE + soft F_1 /MCC	90.8	105.9
Logit adjustment	86.6	91.4
Focal loss	86.3	88.6
OKO	83.4	130.6
CFAL	58.7	108.1
cRT	43.8	47.3

- Nominal patch deprivation costs up to 13.77 pp on BRACS and 4.96 pp balanced accuracy on TCGA-UT
- Matching the weight signal to the deprivation pays off only for patch shortage (+2.3–7.1 pp); for patient-support deprivation it gives no gain (≤ 0.06 pp)
- Scarce-patch allocation matters: depriving easier classes costs 13.23 pp vs. 4.30 pp for harder classes (BRACS)

Difficulty Alignment Hypothesis

Easier classes may have more achievable performance to lose; already-hard classes may have less.

Results

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Distinct-unit support intervention causes largely unrepairable damage

Signal-based Reweighting

Recovery (%)	BRACS	TCGA-UT
Prevalence	0.3	-0.2
Nominal support	0.5	-0.0
Independent support	1.3	-0.5
Difficulty	-5.8	-1.9
Diversity	2.2	-0.3

Wider Roster

Recovery (%)	BRACS	TCGA-UT
Balanced sampling	-21.1	3.8
CE + soft F_1 /MCC	-3.4	6.2
Logit adj. (train/post-hoc)	-0.2	-0.0
Focal loss	-67.1	-0.4
CFAL	-41.7	16.6
OKO	-95.3	13.1
cRT	-34.3	-13.5

- Distinct-unit patient deprivation costs up to 3.03 pp on BRACS and 4.83 pp on TCGA-UT
- This deficit remains largely unrecovered: the best method restores 2.2% on BRACS, against 16.6% on TCGA-UT
- Several methods even decrease performance, up to -95.3% for OKO

Results

Patch deprivation miscalibrates through overconfidence; patient shortage does not

Signal-based Reweighting

NLL recovery (%)	BRACS	TCGA-UT
Prevalence	112	121
Nominal support	86	119
Independent support	-56	26
Difficulty	-11	24
Diversity	-113	10

Wider Roster

NLL recovery (%)	BRACS	TCGA-UT
Balanced sampling	112	133
CE + soft F_1 /MCC	114	127
Focal loss	112	93
OKO	102	83
CFAL	82	-960
Logit adj. (train/post-hoc)	79	66
cRT	-82	-7

- Best methods recover up to 114% (BRACS) and 133% (TCGA-UT) of the NLL loss under severe deprivation, passing the balanced reference
- Little of this is real: one temperature brings every method within ± 0.08 nats of the reference (aggregate NLL)
- Patient shortage: no measurable NLL loss on BRACS; on TCGA-UT balanced sampling stops at 68%, and temperature scaling does not close the rest (27–55% scaled)

What is a nat? The unit of NLL: the surprise of the model at the true class. 0 means certain and right, and +0.1 nats means about 10% less probability on the correct class.

Discussion

What makes distinct-unit-support shortage so damaging?

A USEFUL DECOMPOSITION OF PATCH EMBEDDINGS

$$\underbrace{\mathbf{x}_{gi}}_{\text{patch}} = \underbrace{\mathbf{u}_c}_{\text{true class centre}} + \underbrace{\mathbf{b}_g}_{\text{patient offset}} + \underbrace{\mathbf{e}_{gi}}_{\text{local patch variation}}, \quad \mathbf{b}_g \sim (0, \Sigma_{B,c}), \quad \mathbf{e}_{gi} \sim (0, \Sigma_{W,c})$$

Let $\mathbf{x}_{gi} \in \mathbb{R}^d$ be patch i from patient g .

1. Class-centre error (first moment)

$$\mathbf{e}_{u,c} = \hat{\mathbf{u}}_c - \mathbf{u}_c = \frac{1}{G} \sum_{g=1}^G \mathbf{b}_g$$

Few patients shift the estimated class centre away from its population location.

2. Between-patient covariance error (second moment)

$$\mathbf{E}_{\Sigma,c} = \underbrace{\hat{\Sigma}_{B,c}}_{\text{estimated from sampled patients}} - \underbrace{\Sigma_{B,c}}_{\text{true between-patient covariance}}$$

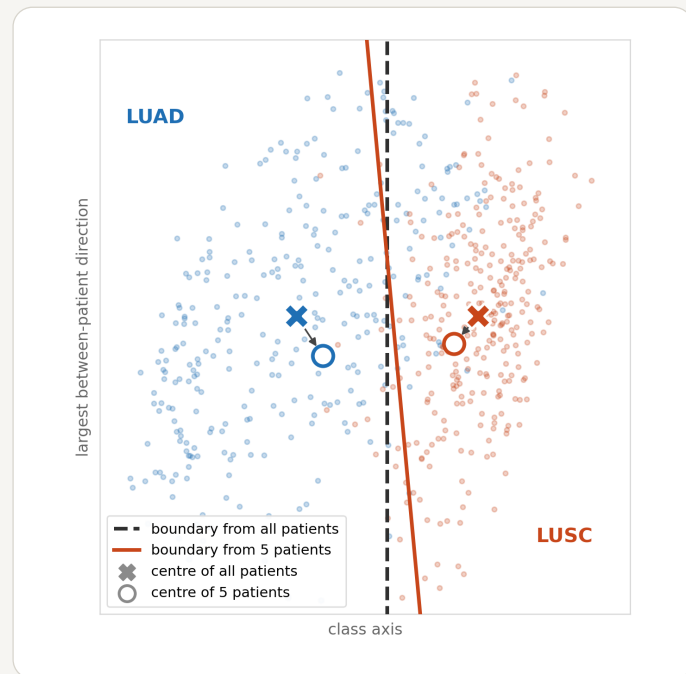
Few patients distort the estimated geometry of patient variation.

A small patient cohort poorly estimates the first two moments of the class-conditional patient distribution.

Discussion

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Class centre error

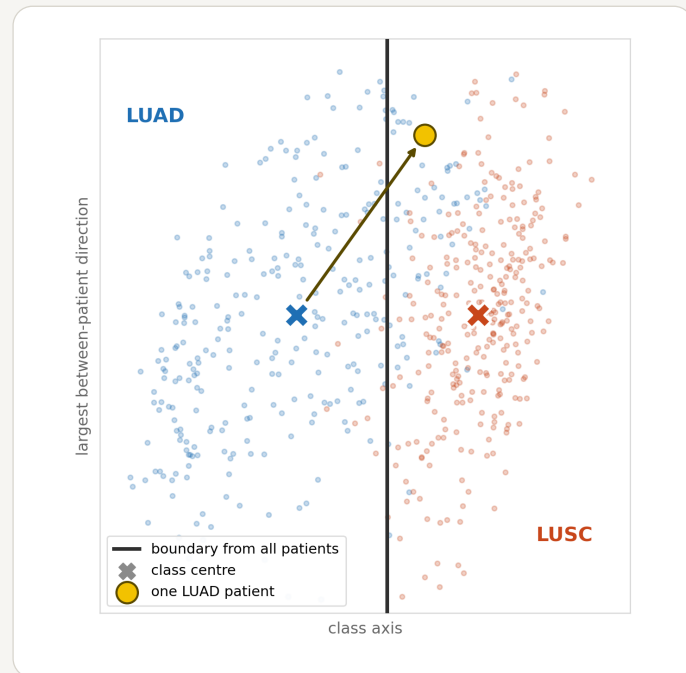


- Few patients estimate each class centre badly: at a fixed patch budget, only more patients shrink it.
- Figure: five patients place the centre ~10 units off a class gap of 16; boundary tilts, error rate 11.8% → 14.4%.
- Correcting only the centres removes 74% of the 5 → 20 patient gap on TCGA-UT (9.41 pp), and four fifths of it on BRACS.
- A patient's value lies in where it places the class: many patients with few patches each beat deep annotation of few patients.

Discussion

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Patient direction error



- Even with correct class centres, a patient carries its own offset b_g along the directions in which patients differ — on TCGA-UT that between-patient term dominates (mean ICC 0.457).
- Figure: 61 of 304 LUAD patients sit on the LUSC side of the boundary; the marked patient is one of them, pushed across by its own offset, not by a wrong centre.
- The classifier weights those directions like any other, so a large offset along them costs as much as real class evidence.
- This is what remains after centre correction: 2.42 pp of the 9.41 pp gap on TCGA-UT.

Discussion

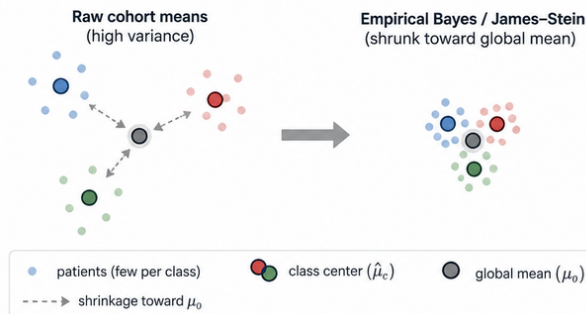
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Potential remedies

Shrinkage of class centers — First moment

Use **Empirical Bayes / James–Stein shrinkage** instead of raw cohort means.

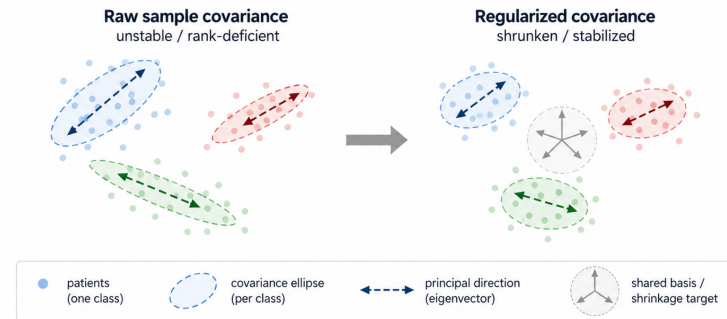
- Directly reduces the dominant class-center estimation variance at small patient counts.
- Strongest motivation: center error explains ~74% of the 5 → 20 patient gap on TCGA-UT and ~82% on BRACS.
- Hypothesis: better $\hat{\mu}_c \Rightarrow$ substantially smaller discrimination gap.



Regularized patient covariance — Second moment

Use **Ledoit–Wolf / OAS shrinkage** instead of raw sample covariance.

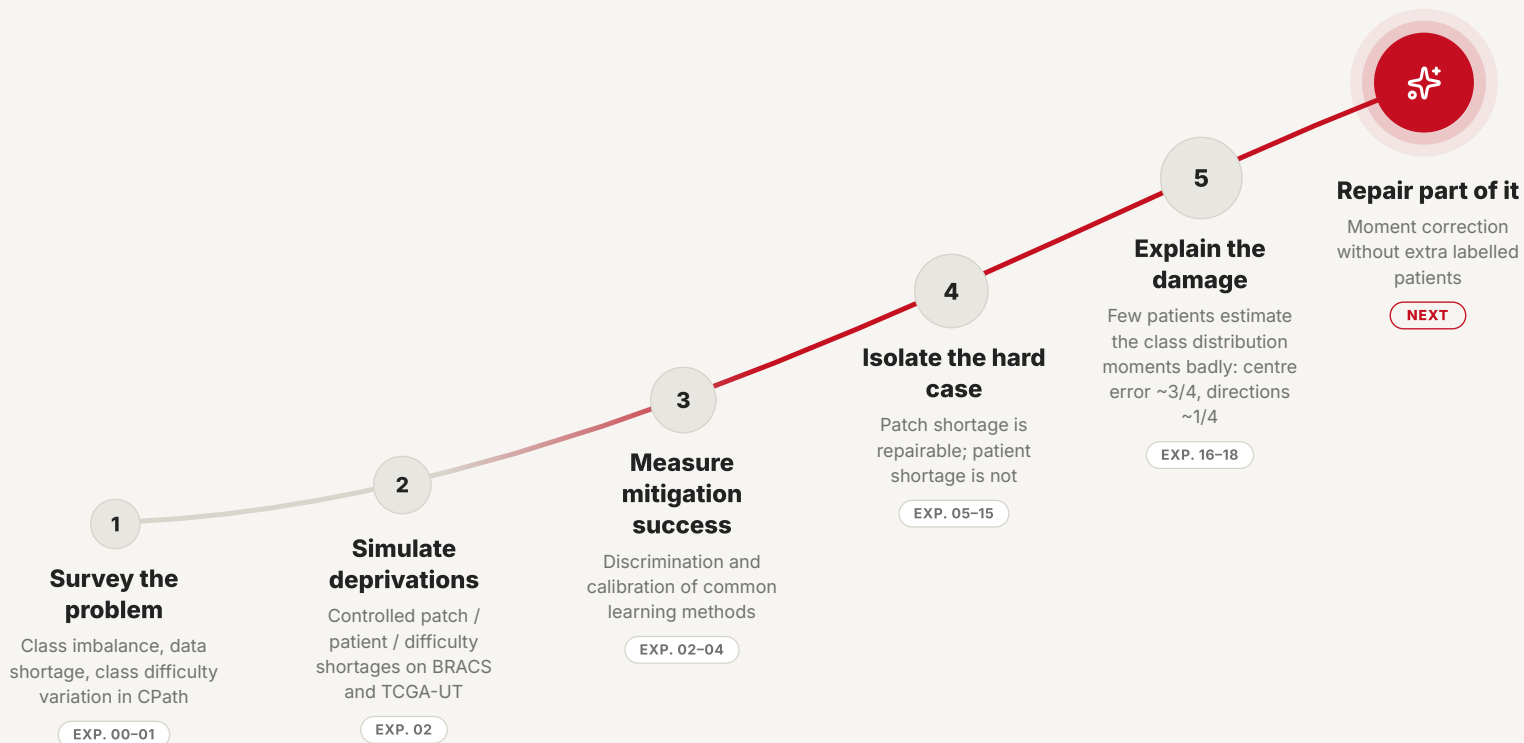
- Stabilizes patient directions and eigenvalues in the $d = 2560$, $G \ll d$ regime.
- Particularly motivated on TCGA-UT; BRACS suggests the independent second-moment contribution is weaker.
- Hypothesis: better $\hat{\Sigma}_{B,c} \Rightarrow$ recovery of the residual gap after center correction.



Summary

Research direction

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Thank you!

Questions and discussion